Your microservice works today… but what happens when its IP changes tomorrow? How will other services find it?

What is Service Discovery?

- Dynamic way services find each other
- Avoids hardcoded IPs and ports
- Essential in distributed systems

Why Service Discovery is Needed?

- Services scale up and down dynamically
- Containers and pods get new IPs
- Manual tracking does not scale

How Service Discovery Works

- Service registers itself
- Discovery system stores location
- Other services query and connect

What is Consul?

- Service registry and discovery tool
- Health checks for services
- Supports service-to-service discovery

What is Kubernetes DNS?

- Built-in service discovery in Kubernetes
- Services communicate using DNS names
- No manual IP management needed

Consul vs K8s DNS

- Consul: External, flexible, multi-platform
- K8s DNS: Native, simple, Kubernetes-focused
- Choice depends on architecture needs

Real-World Use Cases

- Microservices communication
- Load balancing and failover
- Multi-service cloud deployments

Common Mistakes